EXERCISES 3.4

Answers to selected odd-numbered problems begin on page ANS-11.

In Problems 1–10, (**a**) verify that the given function u is harmonic in an appropriate domain D, (**b**) find $v(x, y)$, a harmonic conjugate of u, and (**c**) form the corresponding analytic function $f(z) = u + iv$.

1. $u(x, y) = x$

2. $u(x, y) = 2x - 2xy$

3. $u(x, y) = x^2 - y^2$

4. $u(x, y) = x^3 - 3xy^2$

5. $u(x, y) = \log_e(x^2 + y^2)$

6. $u(x, y) = \cos x \cosh y$

7. $u(x, y) = e^x(x \cos y - y \sin y)$

8. $u(x, y) = -e^{-x} \sin y$

9. $u(x, y) = \dfrac{x}{x^2 + y^2}$

10. $u(x, y) = \arctan(-y/x)$

In Problems 11 and 12, verify that the given function u is harmonic in an appropriate domain D. Find a harmonic conjugate v and find an analytic function $f(z) = u + iv$ satisfying the indicated condition.

11. $u(x, y) = xy + x + 2y - 5$; $f(2i) = -1 + 5i$

12. $u(x, y) = 4xy^3 - 4x^3y + x$; $f(1 + i) = 1 + 4i$

13. (**a**) Show that $v(x, y) = \dfrac{x}{x^2 + y^2}$ is harmonic in a domain D not containing the origin.

(**b**) Find a function $f(z) = u(x, y) + iv(x, y)$ that is analytic in domain D.

(**c**) Express the function f found in part (b) in terms of the symbol z.

14. Suppose $f(z) = u(r, \theta) + iv(r, \theta)$ is analytic in a domain D not containing the origin. Use the Cauchy-Riemann equations (10) of Section 3.3 in the form $ru_r = v_\theta$ and $rv_r = -u_\theta$ to show that $u(r, \theta)$ satisfies Laplace's equation in polar coordinates:

$$r^2 \frac{\partial^2 u}{\partial r^2} + r \frac{\partial u}{\partial r} + \frac{\partial^2 u}{\partial \theta^2} = 0. \tag{5}$$

In Problems 15 and 16, (**a**) use (5) to verify that the given function u is harmonic in a domain D not containing the origin, (**b**) use equations (10) of Section 3.3 to find v, a harmonic conjugate of u, and (**c**) form the corresponding analytic function $f(z) = u + iv$.

15. $u(r, \theta) = r^3 \cos 3\theta$

16. $u(r, \theta) = \dfrac{10r^2 - \sin 2\theta}{r^2}$

Focus on Concepts

17. (**a**) Verify that $u(x, y) = e^{x^2 - y^2} \cos 2xy$ is harmonic in an appropriate domain D.

(**b**) Find a harmonic conjugate v and find an analytic function $f(z) = u + iv$ satisfying $f(0) = 1$. [*Hint*: When integrating, think of reversing the product rule.]

18. Suppose that $f(z)$ is an entire function, $\text{Re}(f(z)) = y^3 - 3x^2y + 2$, and $f(i) = 2 - i$. Find $f(1 + i)$.

19. Express the function f found in Problem 11 in terms of the symbol z.

20. Express the function f found in Problem 16 in terms of the symbol z.

21. (**a**) Show that $\phi(x, y, z) = \dfrac{1}{\sqrt{x^2 + y^2 + z^2}}$ is harmonic, that is, satisfies Laplace's equation $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} + \dfrac{\partial^2 u}{\partial z^2} = 0$ in a domain D of space not containing the origin.

(**b**) Is the two-dimensional analogue of the function in part (a), $\phi(x, y) = \dfrac{1}{\sqrt{x^2 + y^2}}$, harmonic in a domain D of the plane not containing the origin?

22. Construct an example accompanied by a brief explanation that illustrates the following fact:

If v is a harmonic conjugate of u in some domain D, then u is, in general, not a harmonic conjugate of v.

23. If $f(z) = u(x, y) + iv(x, y)$ is an analytic function in a domain D and $f(z) \neq 0$ for all z in D, show that $\phi(x, y) = \log_e |f(z)|$ is harmonic in D.

24. In this problem you are guided through the start of the proof of the proposition:

If $u(x, y)$ is a harmonic function and $v(x, y)$ is its harmonic conjugate, then the function $\phi(x, y) = u(x, y)v(x, y)$ is harmonic.

Proof Suppose $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D. We saw in Section 3.2 that the product of two analytic functions is analytic. Hence $[f(z)]^2$ is analytic. Now examine $[f(z)]^2$ and finish the proof.

Computer Lab Assignments

In Problems 25–28, use a CAS to verify that the given function u is harmonic in an appropriate domain D. In *Mathematica* use the command **D** to find partial derivatives and the command **Simplify** to simplify $\nabla^2 u$.

25. $u(x, y) = \cos\left(x^2 - 2xy - y^2\right) \cosh\left(x^2 + 2xy - y^2\right)$

26. $u(x, y) = \dfrac{1}{2} \log_e \left(\dfrac{x + (1 + y)^2}{x + (1 - y)^2} \right)$

27. $u(x, y) = \cos\left(e^x \sin y\right) \sinh\left(e^x \cos y\right)$

28. $u(x, y) = \sin x \cos y \cosh x \cosh y - \sin y \cos x \sinh x \sinh y$

3.5 Applications

INTRODUCTION In Section 3.4 we saw that if the function $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D, then the real and imaginary parts of f are harmonic; that is, both u and v have continuous second-partial derivatives and satisfy Laplace's equation in D:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \qquad \text{and} \qquad \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0. \tag{1}$$

Conversely, if we know that a function $u(x, y)$ is harmonic in D, we can find a unique (up to an additive constant) harmonic conjugate $v(x, y)$ and construct a function $f(z)$ that is analytic in D.

In the physical sciences and engineering, Laplace's partial differential equation is often encountered as a mathematical model of some time-independent phenomenon, and in that context the problem we face is to *solve* the equation subject to certain physical side conditions called boundary conditions. See Problems 11–14 in Exercises 3.5. Because of the link displayed in (1), analytic functions are the source of an unlimited number of solutions of Laplace's equation, and we may be able to find one that fits the problem at hand. See Sections 4.5 and 7.5. This is just one reason why the theory of complex variables is so essential in the study of applied mathematics.

We begin this section by showing that the level curves of the real and imaginary parts of an analytic function $f(z) = u(x, y) + iv(x, y)$ are two **orthogonal families** of curves.

Orthogonal Families

Suppose the function $f(z) = u(x, y) + iv(x, y)$ is analytic in some domain D. Then the real and imaginary parts of f can be used to